Experiment No: 08

Title: Employee Payroll Using Interface

■ Basic Information

Program File: Week8EmployeePayrollInterface.java

Concepts Used: Interfaces, Implementation, Classes and Objects, Polymorphism, Conditional Operator, User Input

■ Objective / Aim

To develop a Java program using the Payable interface to display salary details for regular and contract employees.

■■ System Requirements

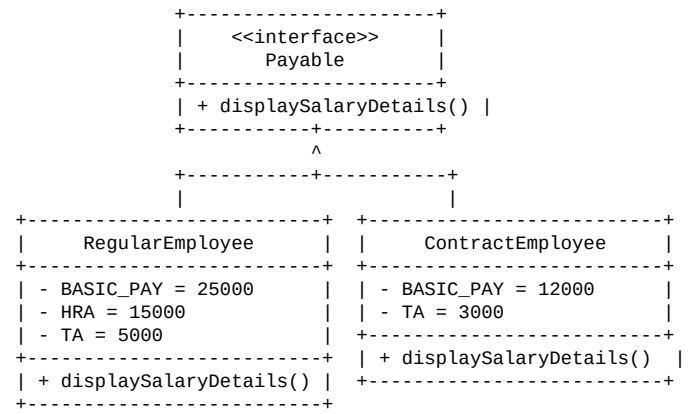
Hardware: Standard PC/Laptop, minimum 4 GB RAM, minimum 500 MB free storage.

Software: Windows/Linux/macOS, Java Development Kit (JDK), Java IDE or Command Prompt/Terminal.

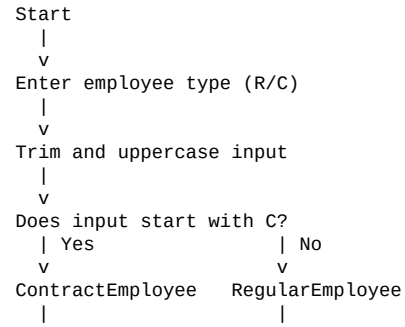
■■ Algorithm / Logic Description

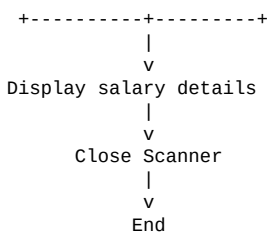
1. Start the program.
2. Declare the Payable interface with the displaySalaryDetails() method.
3. Implement Payable in RegularEmployee and ContractEmployee classes.
4. Set regular employee salary components: Basic Pay = 25000, HRA = 15000, TA = 5000.
5. Set contract employee salary components: Basic Pay = 12000, HRA = 0, TA = 3000.
6. Read the employee type from the user and normalize the input.
7. Create a ContractEmployee if the input starts with C; otherwise create a RegularEmployee.
8. Display the salary details and close the Scanner.
9. Stop the program.

■ UML Class Diagram



■ Program Flow




Display salary details
Close Scanner
End

■ Source Code

```
import java.util.Scanner;

interface Payable {
    void displaySalaryDetails();
}

class RegularEmployee implements Payable {
    private static final int BASIC_PAY = 25000;
    private static final int HRA = 15000;
    private static final int TA = 5000;

    public void displaySalaryDetails() {
        System.out.printf("Basic Pay: %d%nHRA: %d%nTA: %d%nTotal Amount: %d%n",
                BASIC_PAY, HRA, TA, BASIC_PAY + HRA + TA);
    }
}

class ContractEmployee implements Payable {
    private static final int BASIC_PAY = 12000;
    private static final int TA = 3000;

    public void displaySalaryDetails() {
        System.out.printf("Basic Pay: %d%nHRA: 0%nTA: %d%nTotal Amount: %d%n",
                BASIC_PAY, TA, BASIC_PAY + TA);
    }
}

public class Week8EmployeePayrollInterface {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter employee ID (R for Regular, C for Contract): ");
        String type = scanner.nextLine().trim().toUpperCase();
        Payable employee = type.startsWith("C") ? new ContractEmployee() : new RegularEmployee();
        employee.displaySalaryDetails();
        scanner.close();
    }
}
```

■ Compilation & Execution

Step 1: Save the file as **Week8EmployeePayrollInterface.java**.

Step 2: Compile the program using:

```
javac Week8EmployeePayrollInterface.java
```

Step 3: Run the program using:

```
java Week8EmployeePayrollInterface
```

■■ Sample Output

```
Enter employee ID (R for Regular, C for Contract): R
Basic Pay: 25000
HRA: 15000
TA: 5000
Total Amount: 45000

Enter employee ID (R for Regular, C for Contract): C
Basic Pay: 12000
HRA: 0
TA: 3000
Total Amount: 15000
```

■ Result

The program demonstrates the use of a Java interface by allowing regular and contract employee classes to provide their own salary details through the Payable interface.